BCA Semester 1 Examination 2025

Model Solutions: 1BCA1 - Fundamentals of Computer

Session	Paper Code	Time	Full Marks	Pass Marks
2025-2028	111601	3 Hours	100	40

Instructions: Answer any five questions (equal marks). Solutions for all ten questions are given below for practice. Write answers in your own words in the exam.

Q1. Convert the following numbers to decimal

Method: Decimal value = sum of (digit x base^{position}), positions counted from the right starting at 0.

(i) $(11111)_2$

$= 1x2^4 + 1x2^3 + 1x2^2 + 1x2^1 + 1x2^0 = 16 + 8 + 4 + 2 + 1 =$ **31**

(ii) $(1110001)_2$

$= 1x2^6 + 1x2^5 + 1x2^4 + 0 + 0 + 0 + 1x2^0 = 64 + 32 + 16 + 1 =$ **113**

(iii) $(527)_8$

$= 5x8^2 + 2x8^1 + 7x8^0 = 320 + 16 + 7 =$ **343**

(iv) $(EBCD3)_{16}$

Hex values: E = 14, B = 11, C = 12, D = 13.

$= 14x16^4 + 11x16^3 + 12x16^2 + 13x16^1 + 3x16^0$

$= 917504 + 45056 + 3072 + 208 + 3 =$ **965843**

(v) $(BCABBA)_{16}$

Hex values: B = 11, C = 12, A = 10.

$= 11x16^5 + 12x16^4 + 10x16^3 + 11x16^2 + 11x16^1 + 10x16^0$

$= 11534336 + 786432 + 40960 + 2816 + 176 + 10 =$ **12364730**

Number	Decimal
(11111)2	31
(1110001)2	113
(527)8	343
(EBCD3)16	965843
(BCABBA)16	12364730

Q2. What is data representation? Explain UTF-8 encoding system.

Data representation

Data representation is the way data (numbers, text, images, sound, video) is converted into a form a computer can store, process and transmit. A computer understands only two states, ON and OFF, so every kind of data is finally stored as a pattern of binary digits (0 and 1).

- **Numbers:** binary, octal, decimal and hexadecimal systems; integers use sign-magnitude or 2's complement; real numbers use floating point.

- **Characters:** each character is given a numeric code, e.g. ASCII (7/8 bits), EBCDIC, Unicode.
- **Images:** a grid of pixels, each pixel stored as colour values (RGB).
- **Audio / video:** samples of the analog signal converted to digital values (digitisation), often compressed.

UTF-8 encoding

UTF-8 (Unicode Transformation Format - 8 bit) is a variable-length character encoding of Unicode. It represents every Unicode code point using **1 to 4 bytes**. It is the most widely used encoding on the web.

Code point range	Bytes	Byte pattern
U+0000 to U+007F	1	0xxxxxxx
U+0080 to U+07FF	2	110xxxxx 10xxxxxx
U+0800 to U+FFFF	3	1110xxxx 10xxxxxx 10xxxxxx
U+10000 to U+10FFFF	4	11110xxx 10xxxxxx 10xxxxxx 10xxxxxx

Examples

- 'A' = U+0041 fits in 7 bits, so it is stored in 1 byte: 01000001 (0x41), identical to ASCII.
- Euro sign U+20AC needs 3 bytes: 11100010 10000010 10101100 = E2 82 AC.

Features / advantages

- Backward compatible with ASCII (first 128 characters are identical).
- Can represent every character of every language (over a million code points).
- Space efficient for English text (1 byte per character) and no byte-order (endianness) problem.
- Self-synchronising: the start of a character can be identified from the leading bits, so errors do not spread.

Q3. Discuss the following DOS commands: (i) Chdir (ii) Move (iii) Rmdir (iv) Format

(i) CHDIR (or CD) - Change Directory

Type: Internal command. **Purpose:** displays the current directory or changes to another directory.

```
Syntax : CHDIR [drive:][path]   or   CD [drive:][path]
Example: CD \COLLEGE\BCA     (go to BCA inside COLLEGE)
         CD ..               (go up one level)
         CD \                (go to root directory)
```

(ii) MOVE

Type: External command. **Purpose:** moves one or more files to another location, and can also rename a directory.

```
Syntax : MOVE [/Y | /-Y] source destination
Example: MOVE C:\DATA\A.TXT D:\BACKUP        (moves file A.TXT)
         MOVE OLDDIR NEWDIR                   (renames a directory)
```

/Y overwrites existing destination files without prompting; /-Y asks for confirmation.

(iii) RMDIR (or RD) - Remove Directory

Type: Internal command. **Purpose:** deletes a directory. The directory must be empty unless /S is used.

```
Syntax : RMDIR [drive:]path [/S] [/Q]
Example: RD TEST            (removes empty directory TEST)
```

```
RD TEST /S /Q      (removes TEST with all files, no prompt)
```

(iv) FORMAT

Type: External command. **Purpose:** prepares a disk for use by creating tracks, sectors and a file system (FAT/NTFS) and a fresh root directory. **Warning:** all existing data on the disk is erased.

```
Syntax : FORMAT drive: [/FS:filesystem] [/V:label] [/Q]
Example: FORMAT E: /FS:NTFS /V:MYDISK /Q     (quick format of drive E)
```

Q4. Discuss real time and batch processing.

Batch processing

In batch processing, similar jobs/transactions are collected over a period, grouped into a batch and processed together at one time, usually without user interaction. Results are produced after the whole batch finishes.

- Data is collected first and processed later (offline, delayed response).
- Suitable for large volumes of repetitive work; efficient use of CPU, usually run at night.
- **Examples:** payroll, monthly electricity bills, bank statement generation, exam result processing.

Real-time processing

In real-time processing, input is processed immediately as it arrives and the output is produced within a strict time limit (response time is very short, often milliseconds). It may be *hard* real time (missing a deadline is a failure) or *soft* real time (an occasional delay is tolerable).

- Continuous input and immediate output; user/sensor interaction is required.
- **Examples:** air-traffic control, missile guidance, ATM and online ticket booking, patient monitoring, industrial robots.

Comparison

Basis	Batch processing	Real-time processing
Response time	Delayed (hours/days)	Immediate (within deadline)
Data input	Collected and grouped	Processed as it arrives
User interaction	None during processing	Continuous interaction
Cost / complexity	Cheaper, simpler	Costly, complex, needs reliable hardware
Example	Payroll, billing	Flight control, ATM

Q5. Discuss memory hierarchy in a computer system. Differentiate main memory and cache memory.

Memory hierarchy

Memory hierarchy arranges the different types of memory in levels according to speed, cost and capacity. Going down the hierarchy, **speed decreases, capacity increases and cost per bit decreases**. The aim is to give the CPU an average access time close to the fastest memory at a cost close to the cheapest memory. It works because of the *principle of locality* of reference.

Level	Memory	Features
1	CPU Registers	Inside the CPU; fastest, smallest (bytes), most costly

Level	Memory	Features
2	Cache memory (L1, L2, L3)	SRAM; very fast; KB to few MB; holds frequently used data/instructions
3	Main memory (RAM)	DRAM; volatile; GBs; holds running programs and data
4	Secondary storage	SSD, HDD; non-volatile; hundreds of GB to TB; slower
5	Tertiary / offline	Optical discs, magnetic tape, pen drive; cheapest, largest, slowest

Main memory vs cache memory

Basis	Main memory (RAM)	Cache memory
Technology	DRAM (dynamic RAM)	SRAM (static RAM)
Speed	Slower	Much faster
Size	Large (GBs)	Small (KB to MB)
Cost per bit	Lower	Higher
Location	On motherboard (separate chips)	Inside or very close to the CPU
Purpose	Stores programs and data currently in use	Stores most frequently used data to reduce CPU waiting time
Managed by	Operating system	Hardware (cache controller)

Q6. What is DMA? How does DMA work?

Definition

DMA (Direct Memory Access) is a technique in which an I/O device transfers a block of data directly to or from main memory **without continuous involvement of the CPU**. A special hardware unit, the **DMA controller**, manages the transfer, so the CPU is free to do other work.

Main registers of a DMA controller

- **Address register:** starting memory address for the transfer.
- **Word (count) register:** number of words/bytes to be transferred.
- **Control register:** mode and direction of transfer (read or write).

Working (steps)

1. The CPU programs the DMA controller with the memory address, word count and direction, then continues other work.
2. When the device is ready, the DMA controller sends a bus request (HOLD / BR) to the CPU.
3. The CPU finishes its current bus cycle, releases the address, data and control buses and sends an acknowledgement (HLDA / BG).
4. The DMA controller takes control of the bus and transfers data directly between device and memory, incrementing the address and decrementing the count after each word.
5. When the count becomes zero, the DMA controller returns the bus and raises an interrupt to tell the CPU that the transfer is complete.

Modes of DMA transfer

- **Burst (block) mode:** entire block transferred in one go; CPU is blocked meanwhile.

- **Cycle stealing:** controller takes the bus for one word at a time, 'stealing' cycles from the CPU.
- **Transparent mode:** transfer happens only when the CPU is not using the bus.

Advantages: fast transfer of large data (disk, network, graphics), reduced CPU overhead, better system performance.

Q7. Write short notes on any two of the following

(i) Super computer

A supercomputer is the fastest, most powerful and most expensive class of computer, designed for extremely complex calculations. It uses thousands of processors working in parallel and its speed is measured in FLOPS (petaflops/exaflops). **Uses:** weather forecasting, nuclear and space research, genome analysis, climate modelling, AI training. **Examples:** Param Siddhi-AI (India), Summit, Fugaku, Frontier.

(ii) Output devices

Output devices convert the processed result of the computer into a human-readable form. **Soft-copy** output: monitor, speakers, projector. **Hard-copy** output: printers (dot-matrix, inkjet, laser) and plotters. Monitors may be CRT, LCD, LED; printer quality is measured in dpi and speed in pages per minute (ppm).

(iii) Multi-processing

Multiprocessing is the use of two or more CPUs within a single computer system that share memory and peripheral devices and execute different processes at the same time. **Advantages:** higher throughput, more reliability (if one CPU fails the others continue), economy of scale. **Types:** symmetric (SMP) and asymmetric (AMP) multiprocessing.

(iv) MICR

Magnetic Ink Character Recognition is an input technology that reads characters printed with special magnetic ink (iron-oxide based). The MICR reader magnetises the ink and recognises each character from its magnetic signal. It is used mainly on **bank cheques** (cheque number, bank code, account number). **Advantages:** very high accuracy, fast processing, difficult to forge, readable even if the cheque is stamped over. **Limitation:** limited character set and costly equipment.

Q8. Describe the types of operating system with their features.

An operating system (OS) is system software that manages hardware and software resources and provides an interface between user and computer. Main types:

Type	Features	Examples
Batch OS	Jobs with similar needs are grouped and run one after another without user interaction; high throughput, no direct control.	IBM OS/360, early mainframes
Multiprogramming OS	Several programs kept in memory; CPU switches when one waits for I/O, so CPU utilisation is high.	Mainframe systems
Time-sharing / Multitasking OS	CPU time divided into small slices among many users/tasks; quick response, interactive.	UNIX, Windows, Linux
Multiprocessing OS	Uses more than one CPU in one system; parallel execution, reliable and fast.	Windows Server, Linux SMP
Real-time OS (RTOS)	Strict deadlines, immediate response; hard and soft types.	VxWorks, QNX, FreeRTOS

Type	Features	Examples
Distributed OS	Many independent computers appear to the user as a single system; resource sharing, fault tolerance.	LOCUS, Amoeba
Network OS	Runs on a server and lets clients share files, printers and users over a network; central security.	Windows Server, Novell NetWare
Single-user / Multi-user OS	One user at a time (MS-DOS, Windows 10 home) versus many users simultaneously (UNIX).	MS-DOS, UNIX
Mobile / Embedded OS	Light-weight, touch-based or built into devices; power efficient.	Android, iOS, embedded Linux

Q9. Differentiate between any two of the following

(i) Compiler and Interpreter

Compiler	Interpreter
Translates the whole program into machine code at once.	Translates and executes the program line by line.
Generates an object/executable file.	No separate object file is produced.
Execution is faster after compilation.	Execution is slower (translation every run).
Shows all errors after compilation.	Stops at the first error found.
Examples: C, C++.	Examples: Python, BASIC, JavaScript.

(ii) Static and Dynamic Memory (SRAM and DRAM)

Static memory (SRAM)	Dynamic memory (DRAM)
Uses flip-flops (transistors) to store each bit.	Uses a capacitor and a transistor per bit.
Needs no refreshing.	Needs periodic refreshing as the capacitor leaks charge.
Faster, more expensive, low density.	Slower, cheaper, high density.
Used for cache memory.	Used for main memory (RAM).

(iii) SSI and VLSI

SSI (Small Scale Integration)	VLSI (Very Large Scale Integration)
Fewer than about 10 gates (up to ~100 components) on one chip.	Over 100,000 transistors (now millions/billions) on one chip.
Belongs to the 1960s (2nd/3rd generation).	Belongs to the 1980s onward (4th generation).
Used for basic logic gates and flip-flops.	Used in microprocessors, memory chips, and SoCs.
Low speed and high power per function.	High speed, small size, low power.

(iv) CUI and GUI

CUI (Character User Interface)	GUI (Graphical User Interface)
User types text commands.	User uses icons, windows, menus and a mouse.
Commands must be remembered.	Easy to learn and use.
Uses less memory and is faster.	Needs more memory and processing power.

CUI (Character User Interface)	GUI (Graphical User Interface)
Examples: MS-DOS, Unix shell.	Examples: Windows, macOS, Android.

(v) Digital and Analog computer

Digital computer	Analog computer
Works on discrete data in binary (0 and 1).	Works on continuous physical quantities (voltage, temperature, pressure).
High accuracy and general purpose.	Lower accuracy and special purpose.
Stores large amounts of data.	Little or no storage capacity.
Examples: PC, laptop, mobile.	Examples: speedometer, thermometer, seismograph.

Q10. What do you mean by Software? Explain various types of software with examples.

Software

Software is a set of programs, procedures and related documentation that instructs the computer hardware what to do. It cannot be touched (unlike hardware) and is written in programming languages.

1. System software

Software that controls and manages the computer hardware and provides a platform for application software.

- **Operating system:** Windows, Linux, macOS, Android.
- **Language translators:** compiler, interpreter, assembler.
- **Utility software:** antivirus, disk defragmenter, backup, file compression (WinZip).
- **Device drivers:** printer driver, graphics driver.
- **Firmware:** BIOS/UEFI stored in ROM.

2. Application software

Software designed to perform specific tasks for the user.

- **General-purpose (packaged) software:** MS Word, Excel, PowerPoint, Photoshop, web browsers.
- **Special-purpose (customised/tailor-made) software:** payroll system, railway reservation system, hospital management, library software.
- **Open-source / freeware / shareware / proprietary** are classes based on licensing: LibreOffice (open source), Adobe Reader (freeware), WinRAR (shareware), MS Office (proprietary).

Basis	System software	Application software
Purpose	Runs and manages the computer	Performs user tasks
Dependency	Independent, installed first	Depends on system software
Examples	OS, compiler, drivers	MS Word, Tally, games